CO5 AT3 CLOUD:

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Mock Interview – Big Data

1. Explain the difference between HDFS and traditional file systems.

Answer:
HDFS is a distributed file system designed to store very large datasets across multiple machines. Traditional file systems usually store data on a single machine or storage system.

- **HDFS:** Distributed, scalable, fault-tolerant.
- **Traditional file system:** Usually centralized, limited by one machine's storage and processing capacity.
- HDFS uses **block storage and replication** for reliability.

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2. Why is replication important in distributed storage systems?

Answer:
Replication creates multiple copies of data on different nodes. If one node fails, another copy can still be used.

Benefits:

- Fault tolerance
- High availability
- Data reliability
- Better read performance

For example, HDFS commonly uses a **replication factor of 3**.

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3. How does MapReduce divide work between map and reduce phases?

Answer:
MapReduce divides a large task into smaller parallel tasks.

Map phase: Reads input and converts it into intermediate **key-value pairs**.

Shuffle and Sort: Groups intermediate data according to keys.

Reduce phase: Processes each key and its associated values to produce the final result.

Flow:
Input → Map → Shuffle & Sort → Reduce → Output

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4. What role does YARN play in large-scale Hadoop processing?

Answer:
YARN is Hadoop's **resource management and job scheduling layer**.

It:

- Allocates CPU and memory.
- Schedules applications.
- Manages cluster resources.
- Monitors running tasks.
- Recovers resources when tasks or nodes fail.

This allows different applications such as MapReduce and Spark to share the same Hadoop cluster.

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5. Differentiate NAS and SAN in an enterprise data environment.

NAS	**SAN**
Network Attached Storage	Storage Area Network
Provides file-level storage	Provides block-level storage
Accessed using file protocols	Accessed as storage blocks
Easier to manage	More complex
Suitable for file sharing	Suitable for databases and high-performance applications

In short: NAS provides **files**, while SAN provides **blocks**.

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6. What is the purpose of ETL in a big data pipeline?

Answer:
ETL means **Extract, Transform and Load**.

- **Extract:** Collect data from different sources.

- **Transform:** Clean, validate and convert the data.
- **Load:** Store the processed data in a data warehouse or data lake.

ETL makes data consistent and ready for analytics.

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7. How does Apache NiFi help in data integration workflows?

Answer:
Apache NiFi is a **data-flow and integration tool** that moves and processes data between different systems.

It can:

- Collect data from databases, files and APIs.
- Transform and route data.
- Connect different data sources.
- Monitor data flows.
- Provide data provenance and tracking.

Example:
Database → NiFi → Kafka → HDFS → Analytics

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8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

Common failure points include:

- **DataNode failure:** HDFS automatically recreates missing replicas.
- **NameNode failure:** Use NameNode High Availability with Active/Standby nodes.
- **ResourceManager failure:** Use ResourceManager HA.
- **Network failure:** Use redundant network paths.
- **Task failure:** Hadoop retries failed tasks on available nodes.

Overall: Replication, HA, monitoring and automatic recovery provide fault tolerance.

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9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design:

Data Sources → Secure NiFi/Kafka → Validation → Encrypted Data Lake → Processing → Analytics

Security measures would include:

- **Kerberos/RBAC** for authentication and authorization.
- **TLS** for data in transit.
- Encryption for stored sensitive data.
- Data validation and access control.
- Audit logs and data lineage.
- Data masking for sensitive information.

This provides **secure, controlled and traceable data movement**.

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10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:
I would explain it like this:

"Instead of asking one computer to store and process all our data, Hadoop distributes the work across many computers. When the amount of data grows, we can add more computers to the system. Data is replicated so that hardware failures do not cause data loss, and processing happens in parallel to reduce the time required."

Simple architecture:

Large Data → Distributed Storage → Parallel Processing → Analytics → Business Insights

So, Hadoop provides **scalability, reliability and cost-effective processing of large datasets**.